Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/26/2005
PI Name:	Westheimer, David		
Project Title:	VPCAR Thermal Optimization		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	david.t.westheimer@nasa.gov	**Fax:**	FY 281-483-9167
PI Organization Type:	NASA CENTER	**Phone:**	281-483-2804
Organization Name:	NASA - Johnson Space Center		
PI Address 1:	Mail Code EC2		
PI Address 2:	2101 NASA Road 1		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	08/01/2003	**End Date:**	09/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The Vapor Phase Catalytic Ammonia Removal (VPCAR) is one of the leading water recovery technologies under consideration by the Advanced Life Support Element of AHST. This contribution to VPCAR development consisted of additional fluids and thermal review of the system. A numerical model was built and several gravity sensitive compenents were analyzed. 3D solid models of the Wiped Film Rotating Disk and rotary lobe compressor were generated. Suggested test points were provided to the PI and a thermal representative will take part in future testing at Ames Research Center.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Similar technology is used to desalinate salt water.
Task Progress:	The Advanced Thermal team performed several reviews and analyises on VPCAR. We participated in system testing at Ames Research Center. Some issues with microgravity operation were present with the current system, however they seemed solvable with additional engineering.
Bibliography Type:	Description: (Last Updated: 02/27/2004)